

PRACTICE SHEET

1. The value of $2^{2-\log_2 5}$ is equal to:
(a) 4/5 (b) 5/4
(c) 2/5 (d) 5/2
2. If $y = 2^{1/\log_x(8)}$, then x is equal to:
(a) y (b) y^2
(c) y^3 (d) None of these
3. If a, b and c are in GP, then $\log_{ax} x$, $\log_{bx} x$, $\log_{cx} x$ are in:
(a) GP (b) HP
(c) AP (d) None of these
4. If $2 \log(x+1) - \log(x^2-1) = \log 2$, then x equals to:
(a) 1 (b) 0
(c) 2 (d) 3
5. The number of solutions of $\log_2(x-1) = 2\log_2(x-3)$:
(a) 2 (b) 1
(c) 6 (d) 7
6. The value of $e^{(\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \dots + \log_{10} \tan 89^\circ)}$ is equal to:
(a) 0 (b) 1
(c) e (d) 1/e
7. How many numbers of digits are there in 2^{98} ? (Given that $\log_{10} 2 = 0.30103$)
(a) 98 (b) 99
(c) 30 (d) 29
8. If $\log_{30} 3 = a$ and $\log_{30} 5 = b$, what is the value of $\log_{30} 8$?
(a) $3(1+a+b)$ (b) $3(1-a+b)$
(c) $3(1+a-b)$ (d) $3(1-a-b)$
9. What is the least integral value of $2\log_{10} x - \log_x(0.01)$?
(a) 0 (b) 2
(c) 4 (d) 3
10. What is the value of $\log_{\sqrt{b}} a \log_{(c)^{1/3}} b \log_{(a)^{1/4}} c$?
(a) 12 (b) 24
(c) 1/12 (d) 1/24
11. If $\frac{\log x}{\log 5} = \frac{\log 36}{\log 6} = \frac{\log 64}{\log y}$, what are the values of x and y, respectively?
(a) 8, 25 (b) 25, 8
(c) 8, 8 (d) 25, 25
12. If $(\log_3 x)^2 + (\log_3 x) < 2$, then which one of the following is correct?
(a) $0 < x < \frac{1}{9}$ (b) $\frac{1}{9} < x < 3$
(c) $3 < x < \infty$ (d) $\frac{1}{9} \leq x \leq 3$
13. The value of $7 \log \frac{16}{15} + 5 \log \frac{25}{24} + 3 \log \frac{81}{80}$ is:
(a) 0.3010 (b) 0.3512
(c) 0.412 (d) None of the above
14. The value of $\log \frac{70}{33} + \log \frac{22}{135} - \log \frac{7}{18}$ is:
(Given $\log 2 = 0.3010$, $\log 3 = 0.4771$)
(a) -0.0512 (b) 0.4213
(c) 0.3010 (d) None of these
15. The value of $3^{2-\log_3(4\log 36)}$ is:
(Given $\log 2 = 0.3010$, $\log 3 = 0.4771$)
(a) 1.45 (b) 0.69
(c) 1.2 (d) 3.2
16. If $\log_{10} \sqrt{1+x} + 3 \log_{10} \sqrt{1-x} = \log_{10} \sqrt{1-x^2} + 2$, then the value of x is:
(a) 4 (b) -99
(c) 5 (d) None of these
17. $\log_{25} 625 - \log_{31} 961 + \log_{29} 841 =$
(a) 2 (b) 6
(c) 0 (d) 4
18. If $\log x + \log y = \log(x+y)$ then:
(a) $x = y$ (b) $xy = 1$
(c) $y = \frac{x-1}{x}$ (d) $y = \frac{x}{x-1}$
19. If a, b, c are in GP then $\log_a n$, $\log_b n$, $\log_c n$ are in
(a) A.P. (b) G.P.
(c) H.P. (d) not
20. x triples every second. How will $\log_2 x$ change every second
(a) It will double every second
(b) It will triple every second
(c) It increases by constant amount every second
(d) None of these
21. If $\log_3 \log_3 [\log_3 x] = \log_3 3$, then what is the value of x?
(a) 3 (b) 27
(c) 3^9 (d) 3^{27}
22. What is the value of $\frac{\log_{\sqrt{ab}} H}{\log_{\sqrt{ab}\gamma} H}$?
(a) $\log_{a\beta}(\alpha)$ (b) $\log_{a\beta\gamma}(\alpha\beta)$
(c) $\log_{a\beta}(\alpha\beta\gamma)$ (d) $\log_{a\beta}(\beta)$
23. The value of x for which $\log_9 x - \log_9 \left(\frac{x}{10} + \frac{1}{9} \right) = 1$ is:
(a) 2 (b) 4
(c) 9 (d) 10
24. If $\log_4(x^2+x) - \log_4(x+1) = 2$, then the value of x is:
(a) 2 (b) 4
(c) 5 (d) 16
25. The value of $\left(\frac{1}{\log_3 60} + \frac{1}{\log_4 60} + \frac{1}{\log_5 60} \right)$ is:
(a) 60 (b) 5
(c) 1 (d) 0
26. The value of:

- $\left[\frac{1}{\log(p/q)^x} + \frac{1}{\log(q/r)^x} + \frac{1}{\log(r/p)^x} \right]$ is :
 (a) 3 (b) 2
 (c) 1 (d) 0
27. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, then the value of $\log_{10} (1.5)$ is:
 (a) 0.7161 (b) 0.1761
 (c) 0.7116 (d) 0.7611
28. Given $\log 2 = 0.3010$, the number of digits in 5^{20} is:
 (a) 14 (b) 16
 (c) 18 (d) 25
29. If $x = 27$ and $y = \log_3 4$, then $x^y = ?$
 (a) $1/16$ (b) $3/7$
 (c) 16 (d) 64
30. If $x^{18} = y^{21} = z^{28}$, then $3, 3\log_y x, 3\log_z y, 7\log_x z$ are in:
 (a) A.P. (b) G.P.
 (c) H.P. (d) none of these
31. If $\log_3 x + \log_9 x^2 + \log_{27} x^3 = 3$, then the value of x is:
 (a) 6 (b) 3
 (c) 1 (d) 2
32. If $\log_{10} (x - 15) + \log_{10} 10 = 4$, then the value of x is:
 (a) 785 (b) 985
 (c) 1015 (d) None

ANSWER KEY

1.	a	2.	c	3.	b	4.	d	5.	b	6.	b	7.	c	8.	d	9.	c	10.	b
11.	b	12.	b	13.	a	14.	a	15.	a	16.	d	17.	a	18.	d	19.	c	20.	c
21.	d	22.	c	23.	d	24.	d	25.	c	26.	d	27.	b	28.	a	29.	d	30.	a
31.	b	32.	c																

Solutions

Sol. 1. (a)

$$2^{2-\log_2 5} = \frac{2^2}{2^{\log_2 5}} = \frac{4}{5}$$

Sol. 2. (c)

taking log both side

$$y = 2^{\frac{1}{\log_x 8}} \Rightarrow \log_x y = \frac{1}{\log_x 8} \times \log_x 2$$

$$\Rightarrow \log_x y = \log_x 2 = 1/3$$

$$\Rightarrow x = y^3$$

Sol. 3. (b)

a, b, c are in G.P. $\Rightarrow b^2 = ac$

taking log both sides $2\log b = \log a + \log c$

i.e. $\log_x a, \log_x b, \log_x c$ are in AP

$\Rightarrow 1 + \log_x a, 1 + \log_x b, 1 + \log_x c$ are in AP

$\Rightarrow \log_x x + \log_x a, \log_x x + \log_x b, \log_x x + \log_x c$ are in AP

$\Rightarrow \log_x ax, \log_x bx, \log_x cx$ are in AP

$\Rightarrow \log_{ax} x, \log_{bx} x, \log_{cx} x$ are in HP

Sol. 4. (d)

$$2\log(x+1) - \log(x^2-1) = \log 2$$

$$\log \frac{(x+1)^2}{x^2-1} = \log 2 \Rightarrow \frac{(x+1)^2}{x^2-1} = 2$$

$$\Rightarrow \frac{x+1}{x-1} = 2 \Rightarrow x = 3$$

Sol. 5. (b)

$$\log_2(x-1) = 2\log_2(x-3)$$

$$\log_2(x-1) = \log_2(x-3)^2$$

$$\Rightarrow x-1 = (x-3)^2 \Rightarrow x-1 = x^2-6x+9$$

$$\Rightarrow x^2-7x+10=0 \Rightarrow x=2, 5$$

but $x=2$ is not in domain so there is only one solution.

Sol. 6. (b)

$$e^{(\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \dots + \log_{10} \tan 89^\circ)}$$

$$\Rightarrow e^{\log_{10} \tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ}$$

$$= e^{\log_{10} 1} = e^0 = 1$$

Sol. 7. (c)

$$\text{No. of digits} = 98 \log_{10} 2$$

$$= 98 \times 0.3010 = 29.49 \approx 30 \text{ digits}$$

Sol. 8. (d)

$$\log_{30} 8 = 3\log_{30} 2 = 3\log_{30} \frac{30}{15}$$

$$\Rightarrow 3\log_{30} \frac{30}{3 \times 5} \Rightarrow 3(\log_{30} 30 - \log_{30} 3 - \log_{30} 5)$$

$$\Rightarrow 3(1 - a - b)$$

Sol. 9. (c)

$$2\log_{10} x - \log_x(0.01)$$

$$\Rightarrow 2\log_{10} x - \log_x(10)^{-2}$$

$$\Rightarrow 2\log_{10} x + 2\log_x 10$$

$$\Rightarrow 2(\log_{10} x + \log_x 10)$$

We know that

$$\Rightarrow x + \frac{1}{x} \geq 2 \text{ [By the concept of AM} \geq \text{GM]}$$

$$\text{So } \log_{10} x + \log_x 10 \geq 2$$

$$\Rightarrow 2\log_{10} x - \log_x 0.01 \geq 4$$

Sol. 10. (b)

$$= \log \sqrt{b} \cdot a \cdot \log \sqrt[3]{c} \cdot b \cdot \log \sqrt[4]{a} \cdot c$$

$$= 2\log_b a + 3\log_c b + 4\log_a c = 24$$

Sol. 11. (b)

$$\frac{\log x}{\log 5} = \frac{\log 36}{\log 6} = \frac{\log 64}{\log y}$$

$$\Rightarrow \log_5 x = \log_6 36 = \log_y 64$$

$$\Rightarrow \log_5 x = 2 = \log_y 64 \Rightarrow x = 25, y = 8$$

Sol. 12. (b)

$$\Rightarrow (\log_3 x)^2 + (\log_3 x) < 2$$

$$\Rightarrow (\log_3 x)^2 + (\log_3 x) - 2 < 0$$

$$\Rightarrow (\log_3 x + 2)(\log_3 x - 1) < 0$$

$$\Rightarrow -2 < \log_3 x < 1$$

$$\Rightarrow 1/9 < x < 3$$

Sol. 13. (a)

$$\Rightarrow 7\log \frac{16}{15} + 5\log \frac{25}{24} + 3\log \frac{81}{80}$$

$$\log \left(\frac{2^4}{3 \times 5} \right)^7 + \log \left(\frac{5^2}{2^3 \times 3} \right)^5 + \log \left(\frac{3^4}{2^4 \times 5} \right)^3$$

$$\Rightarrow \log \left(\frac{2^{28}}{3^7 \times 5^7} \cdot \frac{5^{10}}{2^{15} \times 3^5} \cdot \frac{3^{12}}{2^{12} \times 5^3} \right)$$

$$\Rightarrow \log 2 = 0.3010$$

Sol. 14. (a)

$$\log \frac{70}{33} + \log \frac{22}{135} - \log \frac{7}{18}$$

$$\Rightarrow \log \left(\frac{70 \cdot 22}{33 \cdot 135} \cdot \frac{18}{7} \right) = \log \frac{8}{9} = \log 8 - \log 9$$

$$\Rightarrow 3\log 2 - 2\log 3 = 3(0.3010) - 2(0.4771)$$

$$= -0.0512$$

Sol. 15. (a)

$$3^2 - \log_3(4 \log 36) = \frac{3^2}{3 \log_3(4 \log 36)} = \frac{9}{4 \log 36}$$

$$= \frac{9}{8 \log 6} = \frac{9}{8(\log 2 + \log 3)} = \frac{9}{8 \times 0.7781} = 1.45$$

Sol. 16. (d)

$$\log_{10} \sqrt{1+x} + 3\log_{10} \sqrt{1-x}$$

$$= \log_{10} \sqrt{1-x^2} + 2$$

$$\log (\sqrt{x+1})(\sqrt{1-x})^3 = \log 100\sqrt{1-x^2}$$

$$\Rightarrow \sqrt{1-x^2}(1-x) = 100\sqrt{1-x^2}$$

$$\Rightarrow \sqrt{1-x^2}(1-x) - 100\sqrt{1-x^2} = 0$$

$$= \sqrt{1-x^2} (x+99) \Rightarrow x=1, -1-99$$

but all these values are not in domain.

Sol. 17. (a)

$$\log_{25} 625 - \log_{31} 961 + \log_{29} 841$$

$$= 2 - 2 + 2 = 2$$

Sol. 18. (d)

$$\log x + \log y = \log (x+y)$$

$$\Rightarrow \log(xy) = \log (x+y)$$

$$\Rightarrow xy = x+y$$

$$y = \frac{x}{x-1}$$

Sol. 19. (c)

a, b, c are in GP

then $b^2 = ac$

taking log both sides

$$2 \log_n b = \log_n a + \log_n c$$

so $\log_n a, \log_n b, \log_n c$ are in AP then

$\log_n a, \log_n b, \log_n c$ are in H.P.

Sol. 20. (c)

x become 3x then

logx become log3x or $\log 3 + \log x$

so it will increase by a constant value log3

Sol. 21. (d)

$$\log_3 \log_3 [\log_3 x] = \log_3 3$$

$$\Rightarrow \log_3 [\log_3 x] = 3$$

$$\Rightarrow [\log_3 x] = 27$$

$$x = 3^{27}$$

Sol. 22. (c)

$$\frac{\log \sqrt{\alpha\beta} H}{\log \sqrt{\alpha\beta\gamma} H} = \frac{\log H}{\log \sqrt{\alpha\beta}} = \frac{\log \sqrt{\alpha\beta\gamma}}{\log \sqrt{\alpha\beta}}$$

$$\Rightarrow \frac{1}{2} \log(\alpha\beta\gamma) = \frac{\log(\alpha\beta\gamma)}{\log(\alpha\beta)} = \log(\alpha\beta)(\alpha\beta\gamma)$$

Sol. 23. (d)

Given equation may be written as:

$$\log_9 x - \log_9 \left(\frac{x}{10} + \frac{1}{9} \right) = \log_9 9$$

$$\Rightarrow \log_9 \frac{x}{\left(\frac{x}{10} + \frac{1}{9} \right)} = \log_9 9$$

$$\Rightarrow \frac{x}{\left(\frac{x}{10} + \frac{1}{9} \right)} = 9$$

$$\Rightarrow x = \frac{9x}{10} + 1$$

$$\Rightarrow x = 10$$

Sol. 24. (d)

$$\log_4 (x^2 + x) - \log_4 (x+1) = 2 \log_4 4$$

$$\Rightarrow \log_4 \left(\frac{x^2 + x}{x+1} \right) = \log_4 (4^2)$$

$$\Rightarrow \frac{x(x+1)}{(x+1)} = 16$$

$$\Rightarrow x = 16$$

Sol. 25. (c)

Given Exp.

$$\Rightarrow \log_{60} 3 + \log_{60} 4 + \log_{60} 5$$

$$\Rightarrow \log_{60} (3 \times 4 \times 5) = \log_{60} 60 = 1$$

Sol. 26. (d)

Given Exp.

$$\log_x \left(\frac{p}{q} \right) + \log_x \left(\frac{q}{r} \right) + \log_x \left(\frac{r}{p} \right)$$

$$= \log_x \left(\frac{p}{q} \times \frac{q}{r} \times \frac{r}{p} \right)$$

$$= \log_x 1 = 0$$

Sol. 27. (b)

$$\log_{10} (1.5) = \log_{10} (3/2)$$

$$= \log_{10} 3 - \log_{10} 2$$

$$= (0.4771 - 0.3010) = 0.1761$$

Sol. 28. (a)

$$\log 5^{20} 20 = 20 \log 5$$

$$= 20 \log (10/2)$$

$$= 20 \times [\log 10 - \log 2]$$

$$= 20 \times (1 - 0.3010)$$

$$= 20 \times 0.6990 = 13.98$$

$$\text{Characteristic} = 14$$

$$\text{Number of digits} = 14$$

Sol. 29. (d)

$$x^y = (27)^{\log_3 4} = (4)^{\log_3 27} = (4)^3 = 64$$

Sol. 30. (a)

Let $x^{18} = y^{21} = z^{28} = k$. then

$$18 \log x = 21 \log y = 28 \log z = \log k$$

$$\log x = \frac{\log k}{18}, \log y = \frac{\log k}{21}, \log z = \frac{\log k}{28}$$

$$\Rightarrow 3 \log_y x = \frac{3 \log x}{\log y} = \frac{3 \times 21}{18} = \frac{7}{2},$$

$$\Rightarrow 3 \log_z y = \frac{3 \log y}{\log z} = \frac{3 \times 28}{28} = 4$$

$$\Rightarrow 7 \log_x z = \frac{7 \log z}{\log x} = \frac{7 \times 28}{28} = \frac{9}{2}$$

3, 7/2, 4, 9/2 are in A.P.

Sol. 31. (b)

Using $\log(a^n) x = \frac{1}{n} \log_a x$, we have

$$\log_{3^2} x^2 = \frac{1}{2} \log_3 x^2 = \log_3 x$$

$$\log_{3^3} x^3 = \frac{1}{3} \log_3 x^3 = \log_3 x$$

Given equ. is

$$3 \log_3 x = 3$$

$$\Rightarrow \log_3 x = 1 = \log_3 3$$

$$\Rightarrow x = 3$$

Sol. 32. (c)

$$\log_{10} (x-15) + \log_{10} 10 = 4$$

$$\Rightarrow \log_{10} (x-15) + 1 = 4$$

$$\Rightarrow \log_{10} (x-15) = 3$$

$$\Rightarrow (x-15) = 10^3 = 1000$$

$$\Rightarrow x = 1015$$

NDA PYQ

1. What is $\log \left(a + \sqrt{a^2 + 1} \right) + \log \left(\frac{1}{a + \sqrt{a^2 + 1}} \right)$ equal to?
 (a) 1 (b) 0
 (c) 2 (d) $\frac{1}{2}$
[NDA (I) 2011]
2. What is the value of:
 $\log_{10} \left(\frac{9}{8} \right) - \log_{10} \left(\frac{27}{32} \right) + \log_{10} \left(\frac{3}{4} \right)$
 (a) 3 (b) 2
 (c) 1 (d) 0
[NDA (I) 2011]
3. What is the value of $\log_2 (\log_3 81)$?
 (a) 2 (b) 3
 (c) 4 (d) 9
[NDA (II) 2011]
4. If $\log_{10} 2$, $\log_{10} (2^x - 1)$, $\log_{10} (2^x + 3)$ are three consecutive terms of an AP, then which one of the following is correct?
 (a) $x = 0$ (b) $x = 1$
 (c) $x = \log_2 5$ (d) $x = \log_5 2$
[NDA (II)-2011]
5. What is the value of $2 \log_8 2 - \frac{1}{3} \log_3 9$?
 (a) 0 (b) 1
 (c) 2 (d) $1/3$
[NDA (I) 2012]
6. If $(\log_3 x) (\log_x 2x) (\log_{2x} y) = \log_x x^2$, then what is the value of y ?
 (a) 4.5 (b) 9
 (c) 18 (d) 27
[NDA (II) 2012]
7. What is the value of $\log_y x^5 \log_x y^2 \log_z z^3$?
 (a) 10 (b) 20
 (c) 30 (d) 60
[NDA (I) 2013]
8. What is $\log_{81} 243$ equal to?
 (a) 0.75 (b) 1.25
 (c) 1.5 (d) 3
[NDA (II) 2013]
9. If $\log_{10} 2$, $\log_{10} (2^x - 1)$ and $\log_{10} (2^x + 3)$ are three consecutive terms of an AP, then the value of x is:
 (a) 1 (b) $\log_5 2$
 (c) $\log_2 5$ (d) $\log_{10} 5$
[NDA (I) 2015]
10. If $\log_8 m + \log_8 \frac{1}{6} = \frac{2}{3}$ then m is equal to?
 (a) 24 (b) 18
 (c) 12 (d) 4
[NDA (II) 2015]
11. If $\log_a (ab) = x$, then what is $\log_b (ab)$ equal to?
 (a) $\frac{1}{x}$ (b) $\frac{x}{x+1}$
 (c) $\frac{x}{1-x}$ (d) $\frac{x}{x-1}$
[NDA (I) 2016]
12. If $x + \log_{10} (1+2^x) = x \log_{10} 5 + \log_{10} 6$ then x is equal to
 (a) 2, -3 (b) 2 only
 (c) 1 (d) 3
[NDA (II) 2017]
13. The value of $\frac{1}{\log_3 e} + \frac{1}{\log_3 e^2} + \frac{1}{\log_3 e^4} + \dots$ up to infinite terms:
 (a) $\log_e 9$ (b) 0
 (c) 1 (d) $\log_e 3$
[NDA (II) 2017]
14. If $x + \log_{15} (1 + 3^x) = x \log_{15} 5 + \log_{15} 12$, where x is an integer, then what is x equal to?
 (a) -3 (b) 2
 (c) 1 (d) 3
[NDA (I) 2018]
15. If $n = (2017)!$ then what is $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{2017} n}$ equal to?
 (a) 0 (b) 1
 (c) $n/2$ (d) n
[NDA (I) 2018]
16. What is $\frac{1}{\log_2 N} + \frac{1}{\log_3 N} + \frac{1}{\log_4 N} + \dots + \frac{1}{\log_{100} N}$ equal to:
 (a) $\frac{1}{\log_{100!} N}$ (b) $\frac{1}{\log_{99!} N}$
 (c) $\frac{99}{\log_{100!} N}$ (d) $\frac{99}{\log_{99!} N}$
[NDA (I) 2018]
17. If $0 < a < 1$, the value of $\log_{10} a$ is negative. This is justified by:
 (a) Negative power of 10 is less than 1
 (b) Negative power of 10 is between 0 and 1
 (c) Negative power of 10 is positive
 (d) Negative power of 10 is negative
[NDA-2018(1)]
18. $\log_7 \log_7 \sqrt{7\sqrt{7\sqrt{7}}}$?
 (a) $3 \log_2 7$ (b) $1 - 3 \log_2 7$
 (c) $1 - 3 \log_7 2$ (d) $7/8$
[NDA (II) 2018]
19. What is the value of $\log_9 27 + \log_8 32$?
 (a) $7/2$ (b) $19/6$
 (c) 4 (d) 7
[NDA (II) 2018]
20. If $(0.2)^x = 2$ and $\log_{10} 2 = 0.3010$, then what is the values of x nearest tenth?
 (a) -10 (b) -0.5
 (c) -0.4 (d) -0.2
[NDA (II) 2018]
21. If $f(x) = \log_{10} (1 + x)$, then what is $4f(4) + 5f(1) - \log_{10} 2$ equal to?
 (a) 0 (b) 1
 (c) 2 (d) 4
[NDA (I) 2019]
22. $f(x) = \log_x 10$ is defined in the domain
 (a) $x > 10$ (b) $x > 0$, except $x = 10$
 (c) $x \geq 10$ (d) $x > 0$, except $x = 1$

- [NDA (I) 2019]**
23. If $x^{\log_7 x} > 7$ where $x > 0$, then which one of the following is correct?
 (a) $x \in (0, \infty)$ (b) $x \in (1/7, 7)$
 (c) $x \in (0, 1/7) \cup (7, \infty)$ (d) $x \in (1/7, \infty)$
- [NDA (II) 2019]**
24. What is the value of $\frac{1}{10} \log_5 1024 - \log_5 10 + \frac{1}{5} \log_5 3125$?
 (a) 0 (b) 1
 (c) 2 (d) 3
- [NDA 2020]**
25. If $x = \log_c(ab)$, $y = \log_a(bc)$, $z = \log_b(ac)$, then which one of the following is correct?
 (a) $xyz = 1$
 (b) $x + y + z = 1$
 (c) $(1+x)^{-1} + (1+y)^{-1} + (1+z)^{-1} = 1$
 (d) $(1+x)^{-2} + (1+y)^{-2} + (1+z)^{-2} = 1$
- [NDA 2020]**
26. The value of x , satisfying the equation $\log_{\cos x} \sin x = 1$, where $0 < x < \frac{\pi}{2}$, is:
 (a) $\frac{\pi}{12}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
- [NDA (I) 2021]**
27. If $n = (100)!$ then what is $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{100} n}$ equal to?
 (a) 0 (b) 1
 (c) 2 (d) 3
- [NDA (I) 2021]**
28. If $\log_{10} 2 \log_2 10 + \log_{10} (10^x) = 2$, then what is the value of x ?
 (a) 0 (b) 1
 (c) $\log_2 10$ (d) $\log_5 2$
- [NDA (II) 2021]**
29. If $\log_{10} 2$, $\log_{10} (2^x - 1)$, $\log_{10} (2^x + 3)$ are in AP, then what is x equal to?
 (a) 0 (b) 1
 (c) $\log_2 5$ (d) $\log_5 2$
- [NDA (II) 2021]**
30. If $\log_x a$, a^x and $\log_b x$ are in GP, then what is x equal to?
 (a) $\log_a (\log_a b)$ (b) $\log_b (\log_a b)$
 (c) $\frac{\log_a (\log_b a)}{2}$ (d) $\frac{\log_b (\log_a b)}{2}$
- [NDA – 2023 (1)]**
31. What is the number of solutions of $\log_4(x-1) = \log_2(x-3)$?
 (a) Zero (b) One
 (c) Two (d) Three
- [NDA – 2024 (1)]**
32. For $x \geq y > 1$,
 Let $\log_x \left(\frac{x}{y} \right) + \log_y \left(\frac{y}{x} \right) = k$,
 Then the value of k can never be equal to
 (a) -1 (b) $-\frac{1}{2}$
 (c) 0 (d) 1
- [NDA – 2024 (1)]**
33. If $\log_b a = p$, $\log_d c = 2p$ and $\log_f e = 3p$, then what is $(ace)^{\frac{1}{p}}$ equal to?
 (a) bd^2f^3 (b) bdf
 (c) b^3d^3f (d) $b^2d^2f^2$
- [NDA – 2024 (1)]**
- Direction for next two:**
- Let $p = \sum_{j=1}^n \log_{10} 2^j$ and $q = \sum_{j=1}^n \log_{10} 5^j$
- [NDA – 2025 (2)]**
34. If $p + q = 66$, then which one of the following is correct?
 (a) $n < 7$ (b) $7 < n < 9$
 (c) $9 < n < 12$ (d) $n > 12$
35. If $p + q = 15$, then what is $q - p$ equal to?
 (a) $\log_{10} 2.5$ (b) $5 \log_{10} 2.5$
 (c) $10 \log_{10} 2.5$ (d) $15 \log_{10} 2.5$

ANSWER KEY

1.	b	2.	d	3.	a	4.	c	5.	a	6.	b	7.	c	8.	b	9.	c	10.	a
11.	d	12.	c	13.	a	14.	c	15.	b	16.	a	17.	b	18.	c	19.	b	20.	c
21.	d	22.	d	23.	c	24.	a	25.	c	26.	c	27.	b	28.	b	29.	c	30.	c
31.	b	32.	d	33.	a	34.	c	35.	d										

Solutions

Sol.1. (b)

$$\log(a + \sqrt{a^2 + 1}) + \log\left(\frac{1}{a + \sqrt{a^2 + 1}}\right) = \log 1 = 0$$

Sol.2. (d)

$$\log_{10}\left(\frac{9}{8}\right) - \log_{10}\left(\frac{27}{32}\right) + \log_{10}\left(\frac{3}{4}\right) = \log_{10} 1 = 0$$

Sol.3. (a)

$$\log_2(\log_3 81) = \log_2 4 = 2$$

Sol.4. (c)

$$\begin{aligned} \log_{10} 2, \log_{10} (2^x - 1), \log_{10} (2^x + 3) &\text{ are in AP} \\ 2\log_{10} (2^x - 1) &= \log_{10} 2 + \log_{10} (2^x + 3) \\ \Rightarrow (2^x - 1)^2 &= 2 \cdot (2^x + 3) \\ \Rightarrow (2^x)^2 - 2 \cdot 2^x + 1 &= 2 \cdot 2^x + 6 \\ \Rightarrow (2^x)^2 - 4 \cdot 2^x - 5 &= 0 \\ \Rightarrow (2^x - 5)(2^x + 1) &= 0 \\ 2^x &= 5 \text{ and } 2^x = -1 \\ 2^x &= -1 \text{ is Not possible} \\ x &= \log_2 5 \end{aligned}$$

Sol.5. (a)

$$2\log_8 2 - \frac{1}{3}\log_3 9 = \frac{2}{3}\log_2 2 - \frac{2}{3}\log_3 3 = 0$$

Sol.6. (b)

$$\log_3 x (\log_x 2x) (\log_{2x} y) = \log_x x$$

$$\frac{\log x}{\log 3} \cdot \frac{\log 2x}{\log x} \cdot \frac{\log y}{\log 2x} = 2$$

$$\frac{\log y}{\log 3} = 2 \Rightarrow \log_3 y = 2 \Rightarrow y = 3^2 = 9$$

Sol.7. (c)

$$\begin{aligned} \log_y x^5 \log_x y^2 \log_z z^3 \\ \Rightarrow 5\log_y x \cdot 2\log_x y \cdot 3\log_z z \\ \Rightarrow 30\log_y x \cdot \log_x y \cdot \log_z z = 30 \end{aligned}$$

Sol.8. (b)

$$\log_{81} 243 = \log_{3^4} 3^5 = \frac{5}{4}\log_3 3 = \frac{5}{4} = 1.25$$

Sol.9. (c)

$$\begin{aligned} 2\log_{10} (2^x - 1) &= \log_{10} 2 + \log_{10} (2^x + 3) \\ 2\log_{10} (2^x - 1) &= \log_{10} 2 \cdot (2^x + 3) \\ \Rightarrow (2^x - 1)^2 &= 2 \cdot (2^x + 3) \\ \Rightarrow 2^{2x} - 2 \cdot 2^x + 1 &= 2 \cdot 2^x + 6 \\ \Rightarrow 2^{2x} - 4 \cdot 2^x - 5 &= 0 \\ \Rightarrow (2^x - 5)(2^x + 1) &= 0 \\ 2^x &= 5 \Rightarrow x = \log_2 5 \end{aligned}$$

Sol.10. (a)

$$\begin{aligned} \log_8 m + \log_8 \frac{1}{6} &= \frac{2}{3} \\ \log_8 \frac{m}{6} &= \frac{2}{3} \Rightarrow \frac{m}{6} = 8^{2/3} = 4 \\ m &= 24 \end{aligned}$$

Sol.11. (d)

$$\begin{aligned} \log_a (ab) &= x \Rightarrow \log_a a + \log_a b = x \\ \Rightarrow 1 + \log_a b &= x \Rightarrow \log_a b = x - 1 \\ \Rightarrow \log_b a &= \frac{1}{x - 1} \end{aligned}$$

$$\Rightarrow \log_b a + \log_b b = \frac{1}{x - 1} + 1$$

$$\Rightarrow \log_b (ab) = \frac{x}{x - 1}$$

Sol.12. (c)

$$\begin{aligned} \Rightarrow x + \log_{10} (1 + 2^x) &= x \log_{10} 5 + \log_{10} 6 \\ \Rightarrow x \log_{10} 10 + \log_{10} (1 + 2^x) &= x \log_{10} 5 + \log_{10} 6 \\ \Rightarrow \log_{10} 10^x + \log_{10} (1 + 2^x) &= \log_{10} 5^x + \log_{10} 6 \\ \Rightarrow \log_{10} 10^x \cdot (1 + 2^x) &= \log_{10} 5^x \cdot 6 \\ \Rightarrow 10^x \cdot (1 + 2^x) &= 5^x \cdot 6 \\ \Rightarrow 2^x \cdot (1 + 2^x) &= 6 \\ \Rightarrow 2^{2x} + 2^x - 6 &= 0 \\ \Rightarrow (2^x + 3)(2^x - 2) &= 0 \\ \Rightarrow x &= 1 \end{aligned}$$

Sol.13. (a)

$$\begin{aligned} \frac{1}{\log_3 e} + \frac{1}{\log_3 e^2} + \frac{1}{\log_3 e^4} + \dots \\ = \log_e 3 + \log_e 3 + \log_e 3 + \dots \\ = \left(1 + \frac{1}{2} + \frac{1}{4} + \dots\right) \log_e 3 \\ = 2\log_e 3 = \log_e 9 \end{aligned}$$

Sol.14. (c)

$$\begin{aligned} x + \log_{15} (1 + 3^x) &= x \log_{15} 5 + \log_{15} 12 \\ x \log_{15} 15 + \log_{15} (1 + 3^x) &= x \log_{15} 5 + \log_{15} 12 \\ \log_{15} 15^x + \log_{15} (1 + 3^x) &= \log_{15} 5^x + \log_{15} 12 \\ \log_{15} [15^x (1 + 3^x)] &= \log_{15} 5^x \cdot 12 \\ 3^x \cdot 5^x (1 + 3^x) &= 5^x \cdot 12 \\ 3^{2x} + 3^x - 12 &= 0 \\ \text{Put } 3^x &= y \\ y^2 + y - 12 &= 0 \\ y = 3 \Rightarrow 3^x &= 3 \Rightarrow x = 1 \\ y = -4 \Rightarrow 3^x &= -4 \text{ (not possible)} \end{aligned}$$

Sol.15. (b)

$$\begin{aligned} \frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{2017} n} \\ \log_n 2 + \log_n 3 + \log_n 4 + \dots + \log_n 2017 \\ \log_n 2 \cdot 3 \cdot 4 \cdot \dots \cdot 2017 &= \log_n (2017)! \\ = \log_{(2017)!} (2017)! &= 1 \end{aligned}$$

Sol.16. (a)

$$\begin{aligned} \frac{1}{\log_2 N} + \frac{1}{\log_3 N} + \frac{1}{\log_4 N} + \dots + \frac{1}{\log_{100} N} \\ \log_N 2 + \log_N 3 + \log_N 4 + \dots + \log_N 100 \\ \log_N 2 \cdot 3 \cdot 4 \cdot \dots \cdot 100 &= \log_N (100)! \\ = \frac{1}{\log_{(100)!} N} \end{aligned}$$

Sol.17. (b)

Negative power of 10 is lie between 0 and 1

Sol.18. (c)

$$\begin{aligned} \log_7 \log_7 7^{7/8} &= \log_7 \frac{7}{8} = \log_7 7 - \log_7 8 \\ \Rightarrow 1 - \log_7 8 &= 1 - 3\log_7 2 \end{aligned}$$

Sol.19. (b)

$$\begin{aligned} \log_9 27 + \log_8 32 \\ = \log_3 3^3 + \log_2 2^5 = \frac{3}{2} + \frac{5}{3} = \frac{19}{6} \end{aligned}$$

Sol.20. (c)

$$(0.2)^x = 2$$

taking log both sides

$$\begin{aligned} \Rightarrow x \log 0.2 &= \log 2 \\ x &= \frac{\log 2}{\log 0.2} = \frac{\log 2}{\log 2/10} = \frac{\log 2}{\log 2 - \log 10} \\ &= \frac{0.3010}{0.3010 - 1} = \frac{0.3010}{-0.6990} \approx -0.4 \end{aligned}$$

Sol.21. (d)

$$\begin{aligned} f(x) &= \log_{10}(1 + x) \\ 4f(4) + 5f(1) - \log_{10} 2 \\ \Rightarrow 4\log_{10} 5 + 5\log_{10} 2 - \log_{10} 2 \\ \Rightarrow 4\log_{10} 5 + 4\log_{10} 2 &= 4\log_{10} 10 = 4 \end{aligned}$$

Sol.22. (d)

Base of log cannot be 1

Sol.23. (c)

$$\begin{aligned} x^{\log_7 x} &> 7 \\ \text{taking log both sides} \\ \Rightarrow \log_7 x^{\log_7 x} &> \log_7 7 \end{aligned}$$

$$\begin{aligned} \Rightarrow (\log_7 x)^2 &> 1 \\ \Rightarrow \log_7 x < -1 \text{ or } \log_7 x > 1 \\ \Rightarrow x < -1/7 \text{ or } x > 7 \end{aligned}$$

Sol.24. (a)

$$\begin{aligned} \frac{1}{10} \log_5 1024 - \log_5 10 + \frac{1}{5} \log_5 3125 \\ = \frac{1}{10} \log_5 2^{10} - \log_5 10 + \frac{1}{5} \log_5 5^5 \\ = \frac{10}{10} \log_5 2 - \log_5 10 + \frac{5}{5} \log_5 5 \\ = \log_5 2 - \log_5 10 + \log_5 5 = \log_5 \frac{2 \times 5}{10} = 0 \end{aligned}$$

Sol.25. (c)

$$\begin{aligned} \frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} \\ = \frac{1}{1+\log_c ab} + \frac{1}{1+\log_a bc} + \frac{1}{1+\log_b ac} \\ = \frac{1}{\log_c c + \log_c ab} + \frac{1}{\log_a a + \log_a bc} + \frac{1}{\log_b b + \log_b ac} \\ \Rightarrow \frac{1}{\log_c abc} + \frac{1}{\log_a abc} + \frac{1}{\log_b abc} \\ \Rightarrow \log_{abc} c + \log_{abc} b + \log_{abc} a \\ \Rightarrow \log_{abc} abc = 1 \end{aligned}$$

Sol.26. (c)

$$\begin{aligned} \log_{\cos x} \sin x &= 1 \\ \frac{\log \sin x}{\log \cos x} &= 1 \Rightarrow \log \sin x = \log \cos x \\ \Rightarrow \log \sin x - \log \cos x &= 0 \\ \Rightarrow \log \frac{\sin x}{\cos x} &= \log \tan x = 0 \\ \Rightarrow x &= \frac{\pi}{4} \end{aligned}$$

Sol.27. (b)

$$\begin{aligned} \frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{100} n} \\ \log_n 2 + \log_n 3 + \log_n 4 + \dots + \log_n 100 \\ \log_n 2 \cdot 3 \cdot 4 \cdot \dots \cdot 100 &= \log_{100!} 100! = 1 \end{aligned}$$

Sol.28. (b)

$$\begin{aligned} \log_{10} 2 \log_2 10 + \log_{10} (10^x) &= 2 \\ \frac{\log 2}{\log 10} \cdot \frac{\log 10}{\log 2} + x \log_{10} 10 &= 2 \end{aligned}$$

$$1 + x = 2$$

$$\Rightarrow x = 1$$

Sol.29. (c)

$\log_{10} 2, \log_{10} (2^x - 1), \log_2 (2^x + 3)$ are in AP

$$2\log_{10} (2^x - 1) = \log_{10} 2 + \log_{10} (2^x + 3)$$

$$(2^x - 1)^2 = 2 \cdot (2^x + 3)$$

$$\Rightarrow (2^x)^2 - 2 \cdot 2^x + 1 = 2 \cdot 2^x + 6$$

$$\Rightarrow (2^x)^2 - 4 \cdot 2^x - 5 = 0$$

$$\Rightarrow (2^x - 5)(2^x + 1) = 0$$

$$\Rightarrow 2^x = 5, 2^x = -1$$

$$\text{If } 2^x = 5$$

$$x \log_2 = \log 5$$

$$x = \log_2 5$$

Sol.30. (c)

$\log_x a, a^x$ and $\log_b x$ are in GP

$$(a^x)^2 = \log_x a \cdot \log_b x$$

$$(a^x)^2 = \frac{\log a}{\log x} \cdot \frac{\log x}{\log b}$$

$$(a^x)^2 = \frac{\log a}{\log b} = \log_b a$$

Taking log both side

$$\log(a^{2x}) = \log(\log_b a)$$

$$2x \log a = \log(\log_b a)$$

$$2x = \frac{\log(\log_b a)}{\log a} = \log_a (\log_b a)$$

$$x = \frac{\log_a (\log_b a)}{2}$$

Sol.31. (b)

$$\log_2(x-1) = \log_2(x-3)$$

$$\frac{1}{2} \log_2(x-1) = \log_2(x-3)$$

$$\Rightarrow \log_2(x-1) = 2 \log_2(x-3)$$

$$\Rightarrow \log_2(x-1) = \log_2(x-3)^2 \quad (x-1) = (x-3)^2$$

$$\Rightarrow x-1 = x^2 - 6x + 9 \Rightarrow x^2 - 7x + 10 = 0$$

$$\Rightarrow (x-2)(x-5) = 0$$

$$x = 2 \text{ or } x = 5$$

but $x = 2$ is not possible because domain of logarithmic functions is positive.

so $x = 5$ is only one answer.

Sol.32. (d)

$$\log_x \left(\frac{x}{y} \right) + \log_y \left(\frac{y}{x} \right) = k$$

$$\log_x x - \log_x y + \log_y y - \log_y x = k$$

$$1 - \log_x y + 1 - \log_y x = k$$

$$2 - \log_x y - \log_y x = k$$

$$2 - (\log_x y + \log_y x) = k$$

$$2 - \left(\log_x y + \frac{1}{\log_x y} \right) = k$$

we know that

$$\left(\log_x y + \frac{1}{\log_x y} \right) \geq 2$$

$$-\left(\log_x y + \frac{1}{\log_x y} \right) \leq -2$$

$$2 - \left(\log_x y + \frac{1}{\log_x y} \right) \leq 0$$

so $k \leq 0$

value of k cannot be 1

Sol.33. (a)

$$\log_b a = p$$

$$\text{then } a = b^p$$

$$\log_a c = 2p$$

$$\text{then } c = d^{(2p)}$$

$$\log_e c = 3p$$

$$\text{then } e = f^{(3p)}$$

.....(iii)

$$(ace)^{\frac{1}{p}} = [b^p \cdot d^{2p} f^{3p}]^{\frac{1}{p}}$$

$$(ace)^{\frac{1}{p}} = bd^2 f^3$$

Sol.34. (c)

$$p + q$$

$$\sum_{j=1}^n \log_{10} 2^j + \sum_{j=1}^n \log_{10} 5^j$$

$$\sum_{j=1}^n j \log_{10} 2 + \sum_{j=1}^n j \log_{10} 5$$

$$= \sum_{j=1}^n j \log_{10} 10 = \sum_{j=1}^n j \quad \dots\dots\dots(i)$$

Given that $p + q = 66$

$$\Rightarrow 1 + 2 + 3 + \dots + n = 66$$

$$\Rightarrow \frac{n(n+1)}{2} = 66$$

$$\Rightarrow n = 11$$

Sol.35. (d)

Given that

$$p + q = 15$$

now by equation (i)

$$p + q = \sum_{j=1}^n j = 15 \quad \dots\dots\dots(ii)$$

$$q - p = \sum_{j=1}^n \log_{10} 5^j - \sum_{j=1}^n \log_{10} 2^j$$

$$\Rightarrow \sum_{j=1}^n j \log_{10} 5 - \sum_{j=1}^n j \log_{10} 2 = \sum_{j=1}^n j \log_{10} \frac{5}{2}$$

$$\Rightarrow \log_{10} \frac{5}{2} \sum_{j=1}^n j = \log_{10} 2.5 \times \sum_{j=1}^n j$$

$$\Rightarrow \log_{10} 2.5 \times 15 = 15 \log_{10} 2.5$$

You can access the solutions of PYQ through 3 Youtube video:

[1. logarithm 2011 to 2015 | NDA mathematics previous year questions by Sandeep Brar](#)

[2. logarithm 2016 to 2020 | NDA mathematics previous year questions by Sandeep Brar](#)

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